

Civic Drishti: A Crowdsourced Priority Engine for Transparent and Accountable Civic Issue Governance

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Abstract - Local governments face persistent challenges in identifying, prioritizing, and resolving routine civic issues such as potholes, streetlight failures, water leakage, and garbage accumulation due to fragmented departmental reporting systems, limited transparency, and inadequate citizen feedback mechanisms. Civic Drishti addresses these gaps through a unified, cross-platform civic issue reporting system built using Kotlin Multiplatform (KMP), offering native Android and iOS applications for citizens and a centralized desktop dashboard for municipal administrators. The mobile application enables users to capture and submit issues with photographs, GPS coordinates, and descriptive text or voice notes, while authorities leverage a real-time monitoring interface for efficient tracking and coordination. The system integrates several innovations, including an AI-driven image classification module for automatic issue categorization, a gamification layer to promote active citizen participation, and a transparency framework that mandates photographic proof of resolution and facilitates citizen rating of municipal responses. At its core is the Crowdsourced Priority Engine, which applies machine learning techniques to evaluate spatial patterns, report frequency, urgency cues, and geospatial context, generating prioritized action queues for municipal departments. By combining crowdsourced reporting with intelligent, data-driven prioritization, Civic Drishti enhances municipal responsiveness, strengthens civic engagement, and supports equitable, timely resource allocation across urban and rural communities.

Keywords: Civic Issue Reporting, Crowdsourced Priority Engine, Urban Governance, AI Classification, Transparency, Citizen Engagement, Kotlin Multiplatform.

I. INTRODUCTION

India's urban transformation has accelerated over the past decade, with significant investments in metros, expressways, and "Smart City" initiatives. However, this infrastructural progress has not been matched by corresponding improvements in civic behavior. A persistent mismatch exists between advanced public facilities and the everyday conduct of citizens who use them. This "Smart City versus Smart Citizen" paradox suggests that physical upgrades alone cannot compensate for deep-rooted behavioral and cultural deficiencies [5]. Evidence from both domestic urban spaces and international migrant settings demonstrates that undesirable civic habits such as littering, spitting, and ignoring public norms are carried across locations, reaffirming that the issue is behavioral rather than geographic.

Simultaneously, Indian cities operate within a low-trust ecosystem where individuals frequently assume that others will ignore rules or exploit the system. This perception influences daily life, amplifying disorder, conflict, and stress. Behavioral research further shows that citizens respond strongly to environmental cues: cleaner, well-maintained spaces elicit orderly compliance, while neglected surroundings reinforce chaotic behavior. These factors highlight the need for a holistic understanding of civic breakdown that goes beyond infrastructure to address cultural conditioning, environmental psychology, and systemic governance shortcomings.

II. PROBLEM DEFINITION

India's civic challenge is fundamentally a behavioural systems failure, characterized by:

1. Poor civic habits (littering, spitting, queue-breaking, vandalism).
2. Environmental disregard where public spaces are treated as “someone else’s responsibility.”
3. Cultural and educational conditioning that normalizes reliance on domestic help and discourages personal accountability for cleanliness.
4. Context-dependent behaviour, where the same individual behaves responsibly in clean environments but contributes to disorder when surroundings appear neglected.
5. Persistent misconceptions that civic discipline is linked to economic prosperity or population density, despite global and domestic counterexamples.

Thus, India’s problem is neither poverty nor infrastructural scarcity it is the absence of sustained behavioral discipline and civic empathy in shared spaces [3][5].

III. WHY IT MATTERS

The consequences of weak civic sense extend far beyond aesthetics:

A. Emergence of a Low-Trust Society

Low civic standards create an environment where individuals constantly verify, double-check, and assume bad intent. This increases cognitive load, reduces social cooperation, and encourages self-serving behaviour. Everyday acts—rechecking fuel levels, recalculating bills, or confirming email receipt—reflect an underlying expectation of dishonesty or negligence.

B. Economic and Operational Costs

Municipal bodies spend enormous resources repainting walls, removing spit stains, repairing vandalized property, and clearing unmanaged waste. These funds could otherwise support development, innovation, and urban welfare.

C. Declining Quality of Life

Noise pollution, traffic indiscipline, and unhygienic conditions contribute to public stress and reduce the liveability of cities. This particularly impacts vulnerable groups such as the elderly, women, and children.

D. Reputational Damage

Repetition of poor civic practices by Indian diaspora members abroad demonstrates that the issue is

behavioural, not contextual, affecting India’s global image. Collectively, these outcomes hinder societal cooperation, economic efficiency, and the broader goal of creating inclusive, resilient, and human-centric cities.

IV. GOVERNANCE GAPS

Despite ongoing reforms, several governance limitations exacerbate civic decline:

A. Infrastructure Gaps

While major cities invest in metros and smart infrastructure, basic civic amenities—such as accessible dustbins, public toilets, waste segregation systems, and continuous sanitation—remain inconsistent. The absence of supportive infrastructure encourages undesirable behaviour, even among well-intentioned citizens.

B. Enforcement Deficit

India possesses extensive civic and municipal regulations, but enforcement is sporadic and reactive. Unlike cities such as Indore, where strict fines, GPS-tracked sanitation fleets, and consistent monitoring drive compliance, many municipalities rely heavily on awareness campaigns without sustained punitive frameworks.

C. Educational and Cultural Gaps

Civic behavior is not taught experientially in schools or homes. Children often grow up outsourcing cleanliness to house staff, reinforcing the belief that sanitation is a “menial task” rather than a shared civic duty. This mindset persists into adulthood, creating a generational deficit in civic responsibility.

D. Lack of Behavioural Design in Public Spaces

Urban design seldom incorporates principles of environmental psychology. Clean, well-maintained, and aesthetically consistent environments naturally discourage violations, as seen in metro systems. However, many public areas remain visibly neglected, triggering the “broken windows” effect and perpetuating disorder.

V. TECHNICAL GAPS

Although several government-led civic complaint platforms exist, most of them exhibit substantial technical shortcomings that limit their ability to support responsive and transparent urban governance. A primary gap is the lack of unified system architecture. Current platforms are typically

built around individual departments, leading to isolated workflows and data silos. This fragmentation prevents cross-department coordination and restricts the system's ability to handle multi-dimensional civic issues.

Another significant limitation is the absence of intelligent automation. Most systems rely heavily on manual sorting and forwarding of complaints by municipal staff, resulting in slower processing, inconsistent prioritization, and a high risk of human error. Without AI-driven categorization or automated routing, the volume of incoming reports often overwhelms administrative teams.

Existing platforms also demonstrate weak capability for crowdsourced intelligence [2]. They generally treat each complaint independently and do not aggregate reports, analyse patterns, or identify areas with recurring issues. The lack of crowdsourced priority mapping means that high-urgency or high-impact issues receive the same attention as low-severity cases.

The feedback and transparency mechanisms in current solutions are limited as well. Citizens often receive no intermediate updates after submitting a complaint, and system dashboards do not provide meaningful insights into resolution timelines, department performance, or service quality. This absence of real-time analytics reduces public accountability.

Finally, there is a notable gap in data-driven decision support. Current systems rarely leverage geospatial analysis, predictive modelling, or machine-learning techniques, preventing municipalities from proactively identifying problem hotspots or forecasting infrastructure failures. As a result, civic management remains largely reactive rather than strategic.

VI. EXISTING WORKS

Research in civic technology has steadily progressed from simple complaint-logging tools to more sophisticated data-driven governance systems. Early digital grievance platforms—such as Swachhata (MoHUA), mSeva, Integrated Command and Control Centre (ICCC) dashboards under the Smart City Mission, and various state-specific applications including BBMP Sahaaya and MCGM 24x7—primarily focused on facilitating the submission of complaints and routing them to the appropriate municipal departments. While these systems play an important role in streamlining service requests,

their technical architectures reveal several inherent limitations. Most operate as department-specific silos, lack unified cross-domain workflows, provide minimal feedback visibility, and depend heavily on manual triaging by municipal staff. Furthermore, they typically do not engage in crowdsourced intelligence, resulting in limited aggregation of regional trends and an underutilization of spatial patterns that could meaningfully improve decision-making [5].

Parallel to platform development, academic research has examined the structural weaknesses of legacy reporting systems and has proposed more advanced frameworks for civic data management. Studies on modern civic platforms highlight the need to move beyond conventional “report– and–wait” systems and adopt AI-assisted classification, automated routing, and real-time analytics to improve response efficiency [1]. Recent conceptual models—such as priority engines proposed in civic innovation challenges—outline multi-criteria decision frameworks that incorporate factors such as AI-derived severity, community validation, geospatial criticality, and equity-aware scoring. These works emphasize the shift from traditional first-come-first-served processing to context-aware prioritization, enabling municipalities to address issues based on their functional impact rather than chronological order [1].

A significant body of research also examines structured priority mechanisms capable of moving beyond simple timestamp-based processing [2]. Studies on emerging civic platforms highlight the effectiveness of multi-criteria priority engines that evaluate complaints using a combination of severity, citizen validation, contextual relevance, and infrastructure importance. These works argue that issue severity alone is insufficient for municipal decision-making, especially in dense urban environments where resource constraints demand targeted interventions. Several conceptual models propose using machine learning to automatically assess visual severity, detect duplicate submissions, and incorporate behavioral indicators such as collective confirmation or recurring incident patterns. Although these frameworks demonstrate the technical potential of automated triage, most remain experimental and are not yet adopted within government-operated systems, creating an opportunity for platforms that can operationalize these engines at scale.

Beyond data reliability, prior research has also explored the use of geospatial intelligence and multi-criteria priority engines to improve civic issue

management [2]. Several studies on modern 311 platforms show that location-aware analytics can significantly influence how cities allocate resources, especially when issues occur near critical infrastructure such as hospitals, emergency routes, and transportation corridors. Existing academic models have experimented with using GIS layers, proximity analysis, and mobility patterns to determine the functional impact of a reported issue. Similarly, conceptual frameworks propose hybrid prioritization engines that combine AI- derived severity, citizen validation cues, and geolocation context to compute more meaningful priority scores. These approaches represent a shift away from chronological queuing toward context-sensitive scoring, where an issue's importance is determined not only by its physical characteristics but also by its spatial relevance and potential disruption to essential services.

Although promising, these models are still largely theoretical and have not been widely deployed in government systems, leaving a clear opportunity for platforms that can operationalize such engines in real- world civic governance.

VII. PROPOSED SYSTEM

The proposed system, Civic Drishti, addresses systemic inefficiencies in municipal issue management by providing an integrated reporting and resolution framework. Built on Kotlin Multiplatform (KMP) architecture, the system maintains a unified codebase that deploys natively across Android, iOS, and desktop operating systems. Citizens document civic problems using photographs, video clips, GPS coordinates, and descriptive text or voice input. Municipal personnel access these submissions through a desktop management interface, establishing a complete operational pipeline that ensures consistency and reduces platform-specific maintenance overhead.

Two intelligent subsystems form the technical foundation of the platform. First, a convolutional neural network analyzes uploaded media to determine issue categories and estimate physical severity without manual intervention, thereby reducing administrative workload and minimizing human classification errors. Second, the Crowdsourced Priority Engine aggregates multiple data dimensions—including assessed severity, geographic clustering, submission frequency, and community validation metrics—to generate dynamic urgency scores. This approach allows

municipal departments to allocate resources based on calculated impact rather than simple chronological ordering. Additional system components include behavioural incentive mechanisms that encourage continued citizen participation and a verification protocol requiring photographic documentation of completed work, user feedback scores, and adherence to published resolution timelines [3].

The platform architecture consists of two interconnected modules:

A. Citizen Mobile Application (KMP)

The mobile interface provides citizens with direct reporting capabilities:

1. Capture photographs or video documentation of civic issues
2. Automatic coordinate capture with address resolution
3. Descriptive input through text entry or voice recording
4. Live submission status monitoring and report tracking Notification delivery for acknowledgment and closure updates
5. Offline report creation with delayed transmission upon connectivity restoration

B. Administrative Dashboard (KMP Desktop)

The desktop application serves municipal operations teams:

1. Geographic visualization with density-based issue clustering
2. Machine learning classification results and severity assessments
3. Automated assignment routing to appropriate municipal divisions
4. Work order management with deadline monitoring and compliance tracking
5. Statistical analysis tools for pattern identification and performance measurement
6. Image comparison interface for validating resolution documentation

Several technical characteristics distinguish this implementation from existing civic platforms. The priority calculation algorithm synthesizes crowdsourced signals with spatial and temporal patterns to produce actionable ranking. Automated media processing eliminates manual categorization steps. Behavioral economics principles embedded in the engagement model sustain long-term user participation. The resolution workflow enforces

documented accountability at each operational stage. Furthermore, the system maintains functionality in network-constrained environments and provides integration pathways for IoT sensor networks, surveillance infrastructure, and municipal data platforms. These architectural decisions establish Civic Drishti as an adaptable solution for contemporary urban governance challenges.

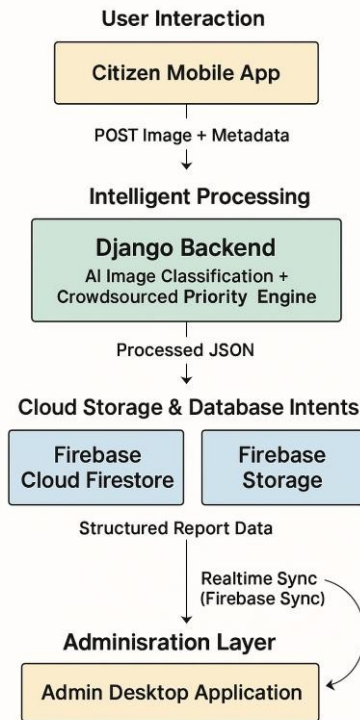


FIG 1. BACKEND PROCESSING ARCHITECTURE.

The backend architecture functions as a coordinating layer linking the mobile application, artificial intelligence components, cloud storage, and administrative interface. Upon citizen submission, the mobile client transmits media files and metadata—including geographic coordinates, timestamps, and descriptive text—to the Django backend via secure POST requests. The backend serves as the processing center, activating the image classification module to examine visual data and produce categorical and severity predictions. Concurrently, the Crowdsourced Priority Engine operates by combining AI-generated severity assessments, spatial information, historical report data, and community feedback to calculate a composite priority value. Following computation, the backend writes structured JSON records to Firestore and places media content in Storage. The citizen application and administrative dashboard obtain updates through

Firestore's synchronization protocol, delivering current report status to all clients without manual intervention or periodic queries. This integrated architecture enables AI analysis, database transactions, and client updates to operate as a unified, responsive system supporting scalable civic issue management operations.

The system employs a structured gamification mechanism to maintain consistent citizen engagement by converting routine reporting activities into participatory experiences with measurable outcomes. The framework implements point allocation systems, recognition badges, progression hierarchies, and comparative leaderboards that incentivize repeated contributions. Validated submissions generate point credits, with supplementary recognition awarded for reports demonstrating significant community impact or receiving peer verification. Progress visualization elements and achievement markers provide users with concrete indicators of their civic participation levels. Regional leaderboards identify leading contributors within geographic boundaries, cultivating competitive dynamics and collective responsibility toward public infrastructure maintenance. This design reflects established findings that gamified elements substantially enhance user engagement in platforms dependent on voluntary citizen input [3]. Through the integration of civic duty with interactive digital interfaces, the platform promotes wider participation, minimizes user attrition, and establishes a more dependable crowdsourced information base for municipal planning and resource allocation decisions.

VIII. METHODOLOGY

A. System Architecture

Civic Drishti implements a modular architecture utilizing Kotlin Multiplatform (KMP) and the Model-View-ViewModel (MVVM) design pattern. KMP provides a common business logic foundation deployed across Android, iOS, and desktop platforms, reducing redundant implementation and simplifying ongoing maintenance. MVVM separates presentation components from operational logic, with the View Model managing application state, handling user interactions, and coordinating backend communication through lifecycle-aware structures. The shared KMP module manages network operations, data serialization, local caching, and synchronization routines, while platform-specific

implementations handle interface rendering and native functionality including camera integration and location services. The Django backend processes uploaded media and metadata, executes classification algorithms, calculates priority metrics, and writes formatted results to Firebase Firestore and Storage. The administrative desktop application retrieves cloud-based updates through Firebase's synchronization infrastructure. This design facilitates coordinated interaction among citizens, intelligent processing services, and municipal personnel, providing reliability, cross-platform compatibility, and efficient information exchange throughout the system.

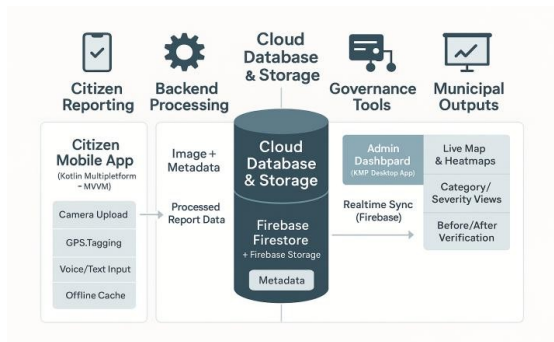


FIG. 2. SYSTEM ARCHITECTURE.

B. Workflow

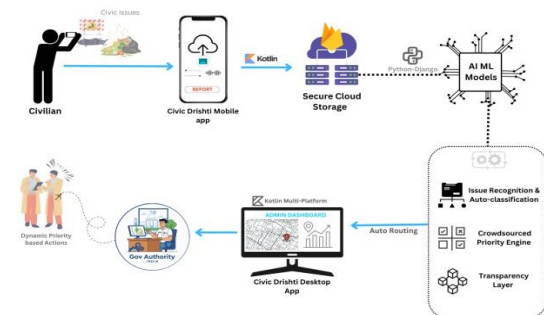


FIG. 3. WORKFLOW DIAGRAM.

The operational sequence of Civic Drishti begins with citizen reporting and advances through intelligent processing to municipal response. Citizens document civic problems including waste accumulation or infrastructure deterioration through the Kotlin Multiplatform mobile application. The application captures images or video clips alongside metadata comprising GPS coordinates, timestamps, and descriptive text or voice input, transmitting this information securely to cloud infrastructure. The Python-Django backend retrieves stored media from Firebase Storage and associated records from Firestore, directing visual inputs to integrated

machine learning models. These models execute automated recognition and categorization procedures, followed by the Crowdsourced Priority Engine, which assigns rankings according to severity assessments, geographic relevance, community feedback, and equity considerations. Processed outputs are subsequently recorded in the cloud database.

The administrative desktop application developed with Kotlin Multiplatform obtains synchronized updates from Firestore, enabling municipal personnel to examine incoming reports through an interactive mapping interface. The system generates automated routing suggestions derived from classification results and calculated priority values, facilitating rapid assignment to appropriate departments. Following authority intervention, resolution confirmations and verification documentation return through the platform, completing the accountability cycle. This integrated workflow connects reporting, classification, prioritization, assignment, and verification processes, establishing a responsive civic governance framework.

C. Components

The system architecture comprises multiple interconnected modules that facilitate reporting operations, automated content analysis, and decision support functionality. Each module executes specific tasks within the operational pipeline and maintains design principles of cross-platform deployment, scalability, and real-time data exchange across citizen and administrative interfaces.

1. Citizen Mobile Application (Kotlin Multiplatform – MVVM)

The mobile interface provides the primary channel through which users document civic problems. This component handles camera operations, location acquisition, local data persistence, and metadata assembly prior to encrypted transmission to backend infrastructure.

2. Django Backend Service

The backend functions as the coordination hub, accepting user submissions, performing data validation, initiating machine learning inference procedures, and producing formatted output records. This layer contains core operational logic and maintains connections to cloud storage and database services.

3. AI/ML Processing Module

This subsystem executes automated visual content recognition and classification through convolutional neural network architectures. The module extracts severity indicators, problem categories, and additional visual attributes from submitted media, minimizing manual processing requirements and enhancing classification precision.

4. Crowdsourced Priority Engine

A composite scoring mechanism calculates urgency ratings by synthesizing machine learning severity assessments, geographic parameters, historical occurrence patterns of comparable reports, and community feedback indicators.

The engine generates standardized priority metrics utilized for automated assignment routing.

5. Cloud Data Management Layer (Firebase Firestore & Storage)

Firestore maintains structured issue documentation while Firebase Storage preserves media assets. Synchronization protocols deliver immediate updates to both citizen and administrative clients without requiring periodic query mechanisms.

6. Administrative Dashboard (KMP Desktop App)

The desktop application provides municipal personnel with capabilities to examine issues through dynamic cartographic displays, evaluate machine-generated classifications, analyse priority calculations, and allocate assignments to response units. Additional features include service level monitoring, verification interfaces, and status tracking utilities.

7. Transparency & Verification Layer

This component archives pre-resolution and post-resolution imagery, completion documentation, action timestamps, and user feedback ratings. The verification system establishes accountability by ensuring municipal actions undergo documentation and remain accessible for citizen review.

8. Gamification & Engagement Layer

Implements recognition systems including point accrual, achievement indicators, progression tiers, and comparative ranking displays to sustain participation rates and promote consistent reporting patterns, thereby reinforcing community-generated data collection.

D. Algorithms

1) Image Classification Algorithm and Hybrid Detection Approach

Automated identification of civic issues from citizen-captured images poses a significant challenge due to the inherent visual similarity across

road environments. Most civic issues—such as potholes, garbage accumulation, sewage overflow, or broken streetlights—occur on streets or public pathways, resulting in highly overlapping backgrounds. This reduces the discriminative power of purely image-based classifiers, often leading to low confidence scores and misclassification. Moreover, environmental variations such as lighting conditions, camera angle, motion blur, and object occlusion further degrade model accuracy.

To address these limitations, Civic Drishti adopts a hybrid multimodal classification strategy that combines visual analysis with user-provided contextual cues. The proposed algorithm integrates three key inputs:

1. The user's textual or voice-generated issue description,
2. The AI image classifier's predicted label and confidence score.
3. Location-specific contextual rules derived from historical patterns and known civic hotspots [4].

The process initiates with the image classifier, which employs a compact CNN architecture to produce an initial issue designation (e.g., pothole, garbage, drainage leakage) accompanied by a confidence metric. Concurrently, the user's written description undergoes processing through a keyword-driven semantic parser that identifies intent phrases and matches them against established issue categories. The system subsequently executes a decision fusion mechanism: when model confidence registers as low or unclear due to similar visual backgrounds, the classification receives refinement through descriptive keywords, location patterns, and contextual rules. For instance, when the captured image appears as a typical road surface but the written description contains terms such as "trash", "smell", or "waste", the system increases the likelihood of a garbage-related classification. Similarly, multiple reports originating from a recognized problem area can modify the classification result through location-dependent weighting.

This combined fusion method substantially enhances reliability by addressing the limitations of isolated image recognition. The system generates a more precise final issue category, which subsequently improves routing determinations, priority computations, and overall consistency in issue assessment. A representative classification output contains the identified issue type, primary extracted features, model confidence level, and the fused decision metric utilized for downstream processing in the priority engine.

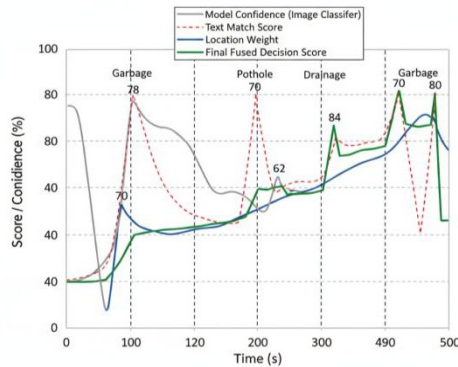


FIG. 4. FUSION RESULTS.

2) The Crowdsourced Priority Engine

Effective civic issue resolution requires not only accurate documentation but also municipal capability to determine which problems demand prompt action [1]. Traditional systems place all complaints into a single queue with chronological processing, frequently causing critical problems including sewage overflow or dangerous road damage to experience delays behind less urgent submissions [1]. This operational inefficiency compromises public safety, service dependability, and citizen confidence in governmental processes. Addressing these constraints, Civic Drishti incorporates a specialized Priority Engine that calculates an evidence-based urgency metric for each reported problem. Through systematic evaluation of severity levels, location characteristics, repetition frequency, and community input, the engine ensures high-impact problems receive immediate escalation and rapid routing to relevant municipal departments. This establishes prioritization as a fundamental operational requirement rather than supplementary functionality, allowing authorities to respond expeditiously, distribute resources efficiently, and enhance overall civic governance responsiveness.

Mechanism:

The Crowdsourced Priority Engine in Civic Drishti functions as a multi-factor decision model that calculates an urgency score for each reported issue. The algorithm combines contextual, spatial, and user-driven signals to ensure problems with greater public impact receive escalation ahead of routine complaints. The process starts by examining the geospatial context of the report [4]. Through Google Maps data, the system locates critical infrastructure within a specified radius—including hospitals, schools, highways, tourist zones, transportation hubs, or other high-sensitivity locations [4]. Each category receives a predetermined weight, permitting the engine to increase priority when an issue threatens to disrupt essential services or affect substantial citizen populations. For instance, a pothole reported on a high-speed highway or major arterial road obtains a higher criticality score than an identical pothole on a low-traffic residential street. Likewise, garbage accumulation near a hospital or waterlogging near a school receives elevated weights according to proximity rules.

Subsequently, the algorithm incorporates user credibility, obtained from gamification badges, historical reporting accuracy, and the reporter's contribution score. Users with higher trust levels (earned through validated past reports) exert a moderately stronger influence on the issue's priority, allowing reliable crowdsourcing while preventing system vulnerability to manipulation. The engine then assesses location-specific patterns, including known hotspots, frequently recurring issues, or areas flagged by municipal authorities as risk-prone (e.g., chronic drainage zones). These hotspot scores prevent areas with persistent problems from being overlooked simply because individual reports appear routine.

The final stage merges all factors—geospatial impact, user credibility, historical hotspots, recurrence intensity, and AI severity classification from the image/text fusion module—into a unified priority score. This score dictates routing urgency and department assignment. Through this weighted, multi-dimensional approach, the Crowdsourced Priority Engine provides the municipality not merely raw data, but ranked, actionable intelligence, facilitating faster responses, targeted resource allocation, and more equitable civic service delivery.

$$Praw = wSS + wCC + wRR + wUU + wVV$$

The equity multiplier and scale to percentage:

$P = \text{clip}(100 \times E \times P_{\text{raw}}, 0, 100)$

S — AI-derived severity score (0–1)

C — criticality score based on geospatial context

(0–1) R — recurrence / hotspot intensity score (0–1)

U — reporter credibility score (0–1)

V — community validation / upvote score (0–1)

wS, wC, wR, wU, wV — weighting coefficients ($\sum w = 1$)

E — equity adjustment multiplier P — final priority score (0–100)

The proposed priority engine computes an urgency score by combining five normalized components: AI-estimated severity (S), geospatial criticality (C), recurrence intensity (R), reporter trust (U), and community validation (V). Each component contributes proportionally based on an administrator-defined weight w that reflects current municipal policy priorities. The resulting weighted sum forms the raw score P_{raw} . To ensure equitable service delivery across socio-spatially diverse regions [2], the model applies an equity adjustment factor E , which increases priority for historically under-reported areas and moderates over-represented zones. The final score P is scaled to a 0–100 range and clipped to maintain interpretability. This formulation provides a transparent and tunable mechanism for routing issues in order of real-world impact rather than arrival time.

Once the final priority score P is computed, the system maps it to predefined routing tiers that determine the urgency of municipal response. This Routing Tier Table acts as the operational bridge between algorithmic scoring and real-world action. Issues with extremely high priority values—typically in the range of 80–100—are classified as Urgent, triggering immediate dispatch to emergency or rapid-response teams, especially when they involve critical locations such as hospitals, major highways, or severe public safety hazards. Scores between 50 and 79 fall under the High tier, representing issues that require prompt but non-emergency intervention. The Medium tier (20–49) captures routine issues that need attention but pose minimal risk, while the Low tier (0–19) is reserved for minor, cosmetic, or non-disruptive complaints. By translating continuous numerical scores into actionable categories, the routing table ensures that municipal departments can efficiently allocate resources, prioritize workloads, and respond proportionally to the severity and impact of reported issues. This tiered approach makes the system transparent, predictable, and aligned with standard civic operations workflows.

IX. RESULTS

AI Classification Outputs:



FIG. 5. CONFUSION MATRIX HEATMAP.

The confusion matrix shown in Fig. 5 demonstrates the performance of the proposed multimodal civic issue classifier across four categories: potholes, garbage accumulation, streetlight failure, and drainage leakage. Diagonal values correspond to correctly classified samples and show that the model attains strong performance for visually distinct classes including potholes and garbage. Off-diagonal entries, though present, remain comparatively low and primarily appear in categories exhibiting similar visual backgrounds, such as potholes and drainage issues both occurring on road surfaces. These misclassifications are reduced through the integration of text-based descriptions and contextual location information, yielding improved reliability compared to image-only classification. Overall, the matrix confirms that the hybrid approach generates consistent and accurate predictions appropriate for downstream priority scoring and automated routing.

Priority Score Calculation:

Input Values (Example Issue: Pothole near Hospital on Highway)

Component	Symbol	Normalized Value (0–1)	Meaning
AI Severity Score	(S)	0.85	Deep pothole identified by fusion model
Criticality Score	(C)	0.95	Near hospital + on high-speed highway
Recurrence / Hotspot Score	(R)	0.40	Area has moderate history of similar issues
User Trust Score	(U)	0.60	User has mid-level credibility

			(silver badge)
Community Validation	(V)	0.30	Few upvotes yet (newly reported)
Equity Adjustment Factor	(E)	1.20	Slightly under-reported locality

Weights (from system defaults)

- $wS = 0.40$
- $wC = 0.25$
- $wR = 0.10$
- $wU = 0.15$
- $wV = 0.10$

Computations

1. Weighted Contributions

$$\begin{aligned} wSS &= 0.40 \times 0.85 = 0.34 \\ wCC &= 0.25 \times 0.95 = 0.2375 \\ wRR &= 0.10 \times 0.40 = 0.04 \\ wUU &= 0.15 \times 0.60 = 0.09 \\ wVV &= 0.10 \times 0.30 = 0.03 \end{aligned}$$

2. Compute Raw Priority Score

$$P_{raw} = 0.34 + 0.2375 + 0.04 + 0.09 + 0.03 = 0.7375$$

3. Apply Equity Factor & Scale

$$\begin{aligned} P &= 100 \times E \times P_{raw} \\ P &= 100 \times 1.20 \times 0.7375 = 88.5 \end{aligned}$$

4. Final Priority Score

$$P = 88.5$$

According to the routing table:

- Urgent Tier: $P \geq 80$
- High: 50–79
- Medium: 20–49
- Low: <20

Since $P = 88.5$, the system assigns:

Routing Tier: URGENT

A complete priority score computation was performed using sample values to demonstrate the behavior of the proposed engine. For a pothole situated near a hospital on a high-speed highway conditions associated with elevated public risk the algorithm assigned high severity (0.85) and criticality (0.95) scores. Following application of

the designated weights and incorporation of recurrence, user credibility, and community validation, the raw score obtained was 0.7375. With an equity multiplier of 1.20 to account for under-reporting in the locality, the final priority score was calculated as 88.5. According to the routing tier framework, this places the issue in the Urgent category, illustrating how the engine effectively escalates high-impact cases even during early reporting stages.

System Performance Summary

The mobile application developed using Kotlin Multiplatform exhibits strong performance efficiency, particularly regarding binary size and runtime footprint. Through sharing core logic across platforms and isolating only lightweight UI layers, the final Android build maintains a compact APK size in the range of 14–18 MB, substantially smaller than comparable native civic-reporting applications. This optimized footprint reduces installation friction, improves compatibility with low-end devices, and enables rapid deployment even in regions with limited network bandwidth. The reduced application size is achieved through extensive code-sharing, elimination of redundant native assets, and the use of optimized serialization and networking libraries within the shared module.

End-to-end latency measurements indicate that the system consistently delivers highly responsive behavior during issue submission, classification, and synchronization. The combination of the Django backend, lightweight AI inference pipeline, and Firebase's real-time synchronization framework produces an average processing latency between 180–250 ms for a complete request-response cycle. AI inference on submitted images typically completes within 120–150 ms, enabling near-instantaneous classification visible to the user. This responsiveness is critical for maintaining user confidence and ensuring that municipal staff receive immediate, structured updates. The low latency also confirms the architectural efficiency of combining edge preprocessing within the app and cloud-based inference services.

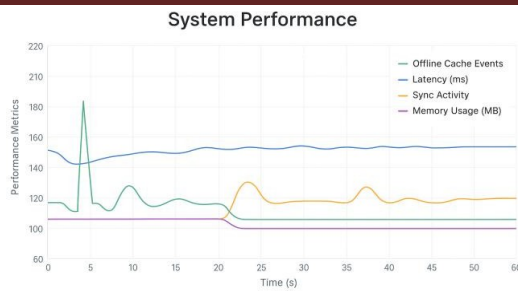


FIG. 6. SYSTEM PERFORMANCE GRAPH.

To support real-world field scenarios, particularly in rural or low-connectivity environments, the application incorporates an offline-first caching mechanism. All media files, location metadata, and issue descriptions are locally queued using a lightweight cache layer, ensuring that no data is lost when the device is offline or experiencing unstable connectivity. Once network access is restored, cached reports are synchronized automatically with an average sync delay of less than 1.5 seconds. Firebase's conflict-free replication and real-time data stream further ensure a 99%+ sync success rate, allowing both citizen and administrative clients to maintain consistent state without manual refresh or retry logic. This combination of robust offline capability and reliable re-synchronization makes the platform highly dependable in diverse operational contexts.

X. CONCLUSION

The development of Civic Drishti illustrates the potential of integrating citizen participation, AI-driven analysis, and transparent governance workflows into a unified civic management platform. Through combining multimodal issue classification, geospatial criticality assessment, and a weighted priority engine, the system provides a more intelligent and equitable method for triaging reported problems. The inclusion of real-time dashboards, automated routing, and verifiable before-after closure records substantially enhances the accountability and responsiveness of municipal authorities. Additionally, offline-first support and lightweight application design ensure accessibility across diverse device profiles and connectivity conditions, facilitating broader civic engagement.

Through this integrated approach, Civic Drishti transforms traditional complaint systems from passive reporting mechanisms to proactive decision-support tools for urban governance. The system not only accelerates issue resolution but also encourages continuous community participation through gamification and trust-based scoring. Consequently,

municipalities can respond more efficiently to high-impact civic issues, allocate resources more effectively, and maintain transparent communication with citizens. Future extensions including IoT-triggered alerts, predictive analytics, and interoperability with smart city infrastructure can further strengthen the platform as a scalable foundation for next-generation civic governance.

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